

Final Project Presentation

Online Attendance & Pay Roll System Through RFID for MIA Enterprises



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Read In the name of your Lord, Who created you...

(IQRA-1, Holy Quran)

TOPICS

1. **INTRODUCTION** ✓
2. **SMART LABELS**
3. **EXISTING SYSTEM**
4. **PROPOSED SYSTEM**
5. **SOFTWARE ARCHITECTURE**
6. **SOFTWARE MODULES**
7. **FUTURE ENHANCEMENT**
8. **Q & A SESSION**

INTRODUCTION

How MIA founded?

Muhammad Bin al-M.Hiba (U.A.E)

Ismael Abdullah

M. Ashraf (Pakistan)

03 Persons were founder of an Engineering Organization “**MIA**” in Middle East (U.A.E).

- Software uses most demanding technology that is RFID.
- Software is only used by authorized person.
- Attendance can be marked with or without RF card.
- A web based login and report viewing.
- Having manual and automatic attendance.
- Computer generated reports & pay checks.

SMART LABELS

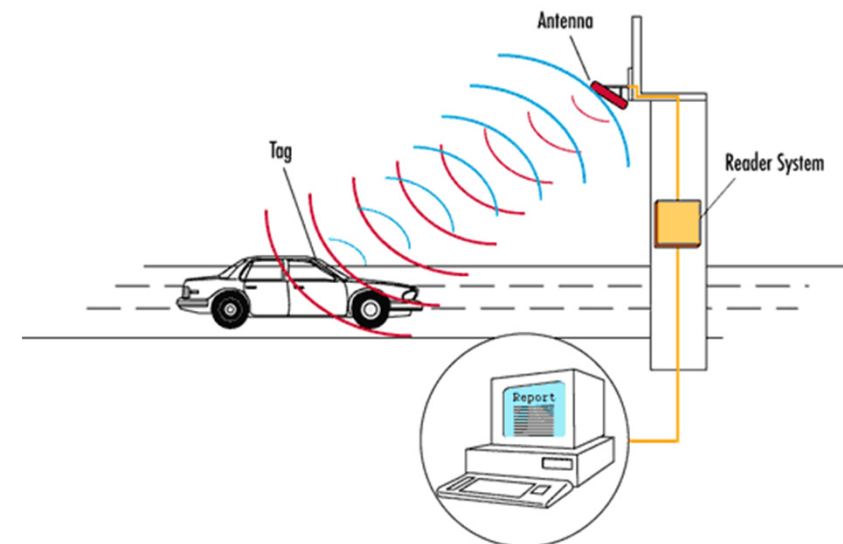
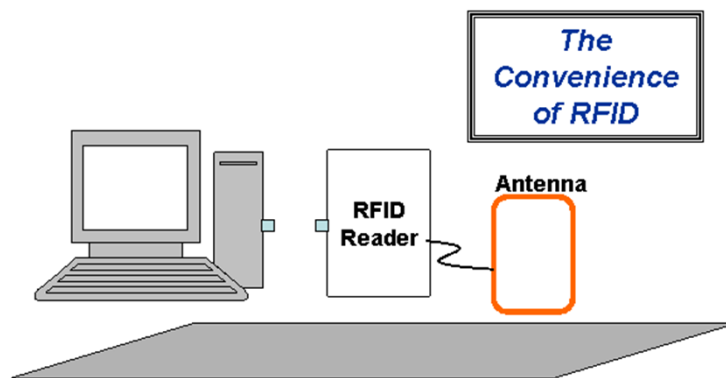
- **What are smart labels?**

- Flat responsive electronic device.
 - Typically no more than 0.3 millimeters thick.
 - Different types of Smart Labels exist in market but 04 types of Smart Labels get more popular.
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- **Electronic Article Surveillance (EAS)**
 - **Smart labels for Brand Enhancement**
 - **Smart labels for diagnostics**
 - **Radio Frequency Identification (RFID)** 

- **RFID** (Main Focus)
- **Components of RFID Kit**

4) What is RFID?

- **R**ADIO **F**REQUENCY **I**DENTIFICATION (**RFID**).
- A wireless communication between objects and RF readers.
- RF Tags are capable of reading, writing, transmitting, storing/updating info.
- Works effectively in hostile environments like excessive dirt, dust or poor visibility
- Most outstanding benefits : Read at remarkable speed App: 1000 Tags per sec.



There are three main components of RFID kit

- **RFID Transponder** (Smart Labels/Cards/Tags)
- **RFID Reader** (Having Internal/External Antenna)
- **RFID Antenna** (Optional)

RFID Transponders

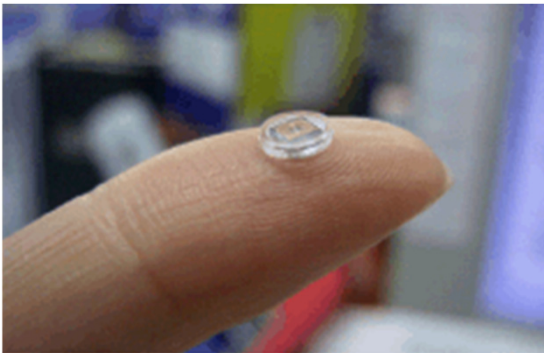
- Small silicon microchip attached to an antenna. Tags holds unique data and other unique attributes of the item.
- Size of Tag: Visiting card to Full Stop or depends on the size of the antenna.

There are Two (02) type of Tags.

1). **Passive Tags** (Main Focus) ✓

2). **Active Tags**

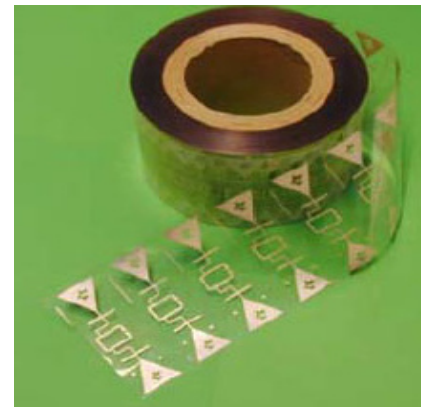
Tag in small size



RF Tags



RF Tag Rolls



Hitachi product Mini size of Tag



1). Passive Tags

- Which are powered solely by the RFID Antenna/reader.
- When the power to the silicon chip on the Tags meets the minimum voltage threshold it require to turn on, the silicon chip.
- Range is usually limited to several meters.
- The amount of data can be **up to 128 bytes** for Passive tags
- Tags are usually made to work at specific frequencies which are licensed free.
- Low Frequency (LF) 125-135 KHz
- High Frequency (HF) 13.56 MHz
- Ultra High Frequency (UHF) 868-930 MHz
- NADRA uses 902-928 MHz freq for VITS

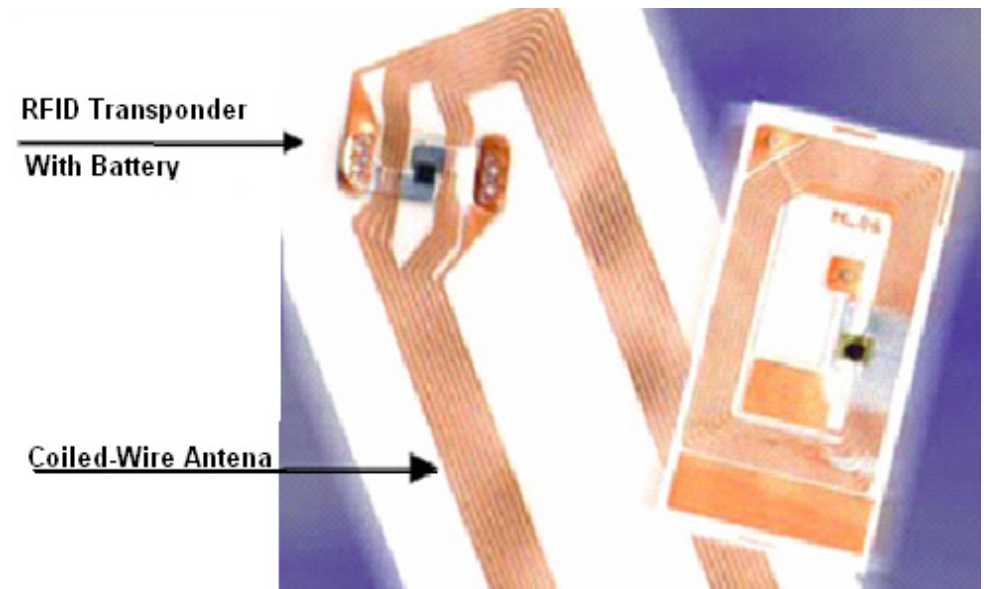


After merging Tags into products



2). Active Tags

- Having their own power source, so they can receive a weaker signal.
 - Its ranges of many tens of meters and even hundreds of meters.
 - The amount of data can be **up to 128 Kbytes** for Active tags
 - Its cost more because of their size and sophistication.
 - Battery life can also limit the life of the Tags.
 - Most common global frequency for active Tags - 433MHz
 - In special situation of very high-speed reading of Rail / Road vehicles (Non-stop tolling).
-
- Ultra High Frequency (UHF) 868-930 MHz
 - Microwave 2.45 GHz & 5.8 GHz



Tags with Chip Vs Without Chip

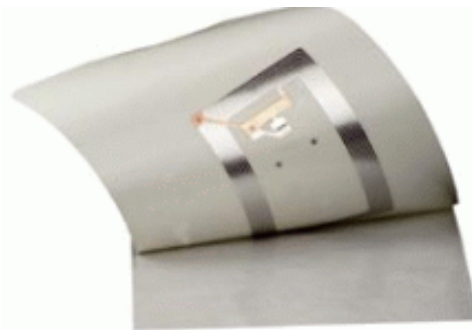
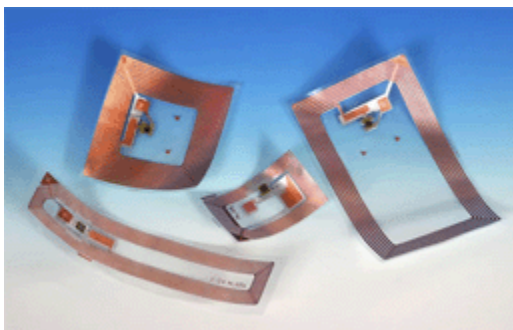
Tags with Chip

- Chip Tags have more cost and more data.
- They can hold larger amounts of data.
- Chip Tags are not usually available below 42 PKR (70 cents) if ordered in limited quantities.
- These uses where data handling is the primary challenge.

Tags without Chip

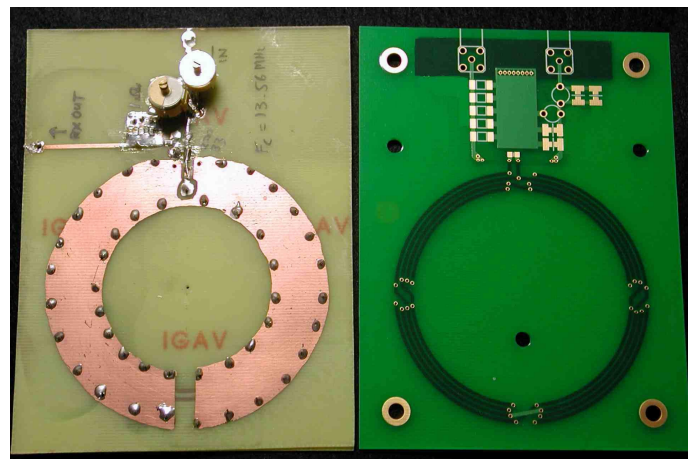


- They are much cheaper but less quantity of data.
- Chipless Tags are usually 20 PKR (30 to 40 cents) even for orders as few.
- Chipless Tags have the price advantages over chip Tags.



RFID Antenna

- Antenna attached to a reader to transmit an electromagnetic fields.
- Electromagnetic field that activates a passive tag when it is within reading range.
- During rewriting applications the antenna of the reader acts as a relay device in the reverse direction.
- The reader communicates a message through its antenna, which Transfers and stores the new data to the activated transducer via its antenna.
- The RFID tag's antenna is practically maintenance free and can be configured in a variety of shapes and sizes ranging in size from a grain of rice to the size of a brick (Zebra Technologies, 2002).



RFID Reader

- May be integrated into handheld device or may be stationary and positioned at strategic points.
- Handheld readers offer portability and stationary devices offer a larger reading range.
- Receives instructions and information from Antenna.
- The Antenna (Scanner's) information is then converted into a digital format by reader.
- And computer can then use for data Analysis, Recording, and Reporting.
- There are Readers today that can simultaneously read 100 to 2000 Tags Per Second.

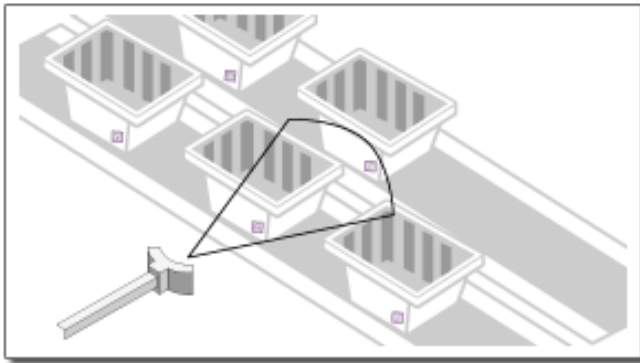


IF5 Fixed "Smart" Reader



Intermec IP3 Portable Reader

RFID VARIOUS APPLICATIONS



Pharmaceuticals/ Food/Hotels
and package (PTA)



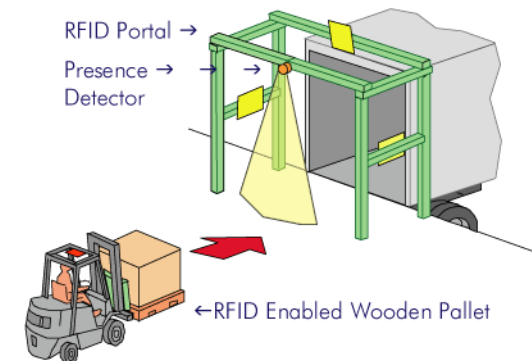
Special Children Schools
(ATA)



Attendance/Pay roll
(PTA)



Special Children Monitoring
(ATA)



**RFID Conceptual Physical Design
for Warehouse Manaaement**

TOPICS

1. INTRODUCTION
2. SMART LABELS
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Existing System of MIA

- Healthy amount of staff Approx: 450.
- The existing system is Manual System.

- **The Manual System**

- **The Registers/Filing System**

- Different departments manages their own Attendance.
- Both System are inefficient and have considerable drawbacks.
- System is unstable to fulfill the growth and need of MIA operations.
- According to organization need, it confuses about their Employee availability.

Draw backs in Existing System

Listed below few drawbacks in Existing system

- Wastage of Resources and Time.
- Neither user friendly nor detailed/online supported.
- Duplication and inconsistency of data
- The data input is laborious and inaccurate
- No data backup for security
- Sharing of data is very difficult
- Very weak data security and privacy
- Report generation is time consuming and laborious
- No urgent reports

Problem Definition

1. Need of a system that Reduce the Dependency on Manual/File/cabin/Registers System.
2. Need of an efficient system to meet the growing requirements in HR Department.
3. Organization want to manage its standard/upgrade its employee status via easy support/queries/feedbacks and etc.
4. Manual and filing based office management system have consider drawbacks that waste the resources
5. Organization want to Share/Present its work with their Employees.
6. Organization want to Present an Online friendly system, but the current system cannot do so.
7. There is no way, That Employee check/access his/her Monthly Attendance/Performance/Pay details.
8. Organization want to manage its standard/upgrade its employee status via easy support/queries/feedbacks and etc.

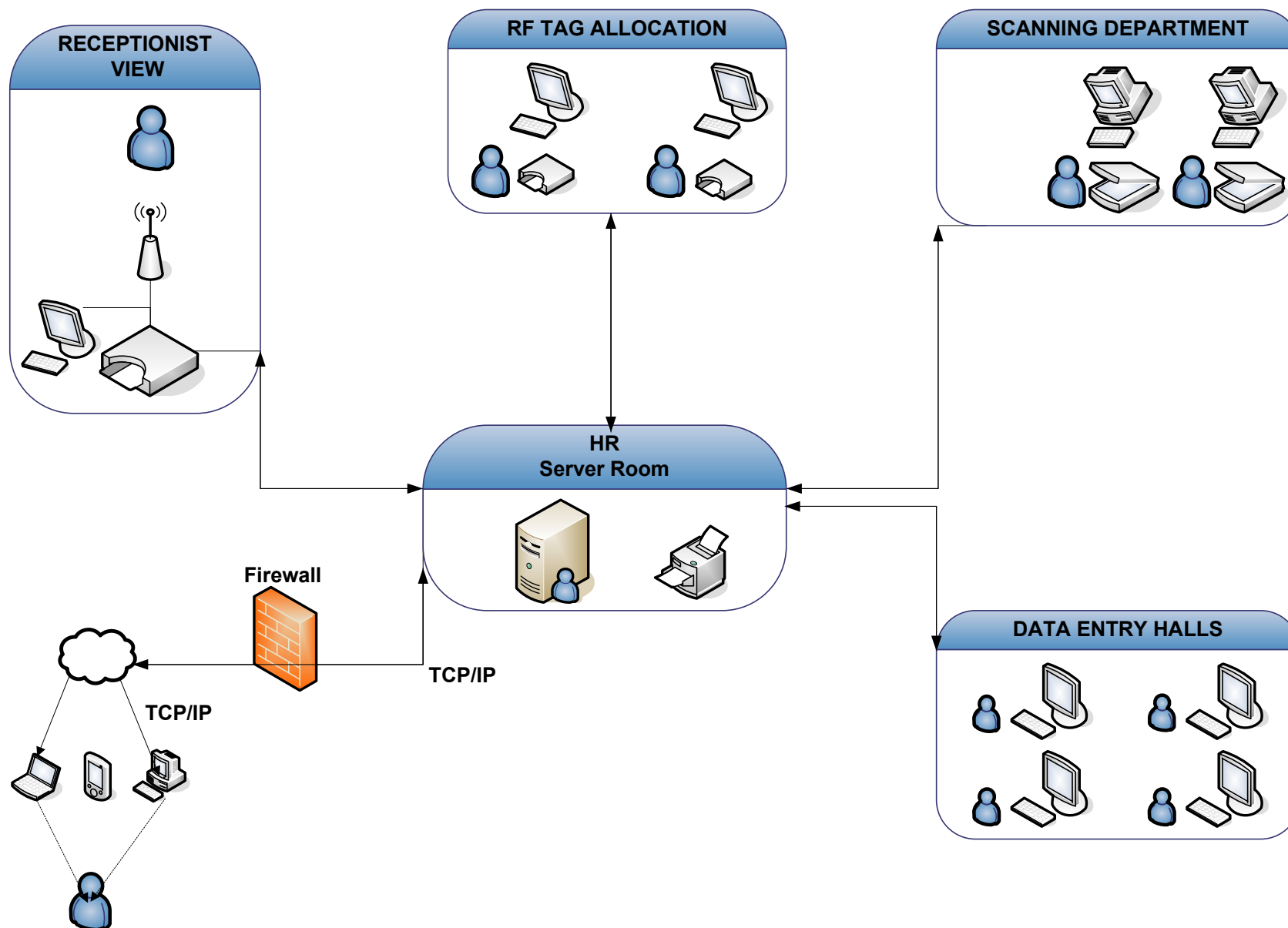
Proposed System (Overview)

- Proposed System is the database approach based on the latest technology.
- A Long life very cheap Modern technology RF based Solution.
- Attendance Based on Smart labels (RFID Cards) that is latest technology.
- Provides solutions of most problems those exist in existing system.
- ORACLE is a reliable and efficient RDBMS in the market.
- Dot NET Frame Work C# 2005 Developer environment enables us to develop user friendly interface that is highly compatible with the database.
- Web Pages provides its Employees to satisfaction/easy feedback/information sharing tool.
- Also Provide **Employee Tracking facility**.

Proposed System (Salient Features)

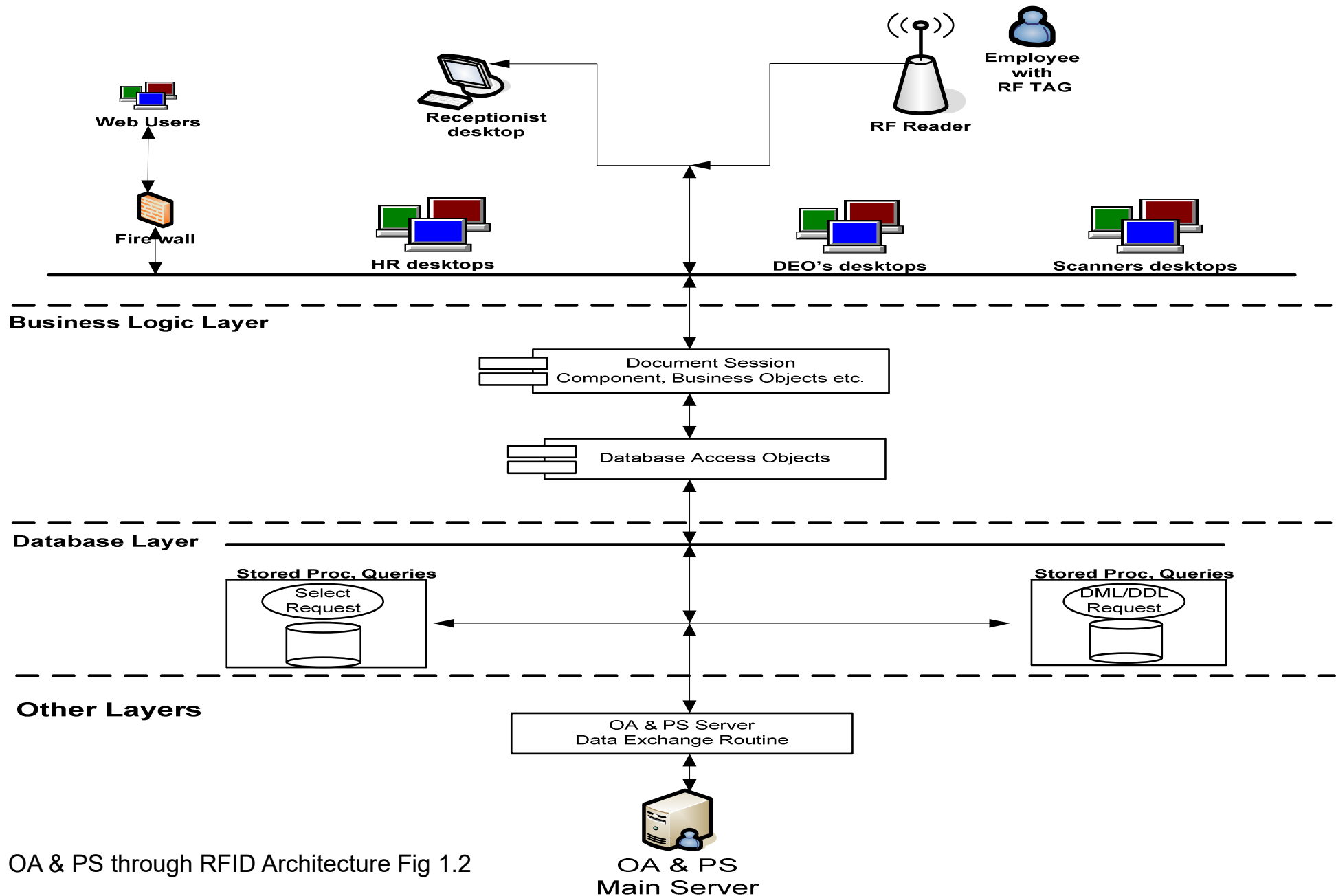
- Newly Automated RF Application for Attendance and Pay roll system
- Radio Waves are not harmful for human health
- Time and Resources saving system
- Minimal data redundancy, consistency and Integration of data with minimum storage space.
- Sharing of data, enforcement of standards
- Ease of application development
- Uniform security, Privacy, Reduced program maintenance
- Facilitated, and accurate data input
- Monthly, yearly detailed reports
- A Complete /Stable/Automated Help for HR also supported to Web based views.
- Employee facility to logon on Organization Website to check monthly details
- Also support to Employee Detailed IN/OUT (Tracking) times.

The Proposed System



OA & PS through RFID Proposed System Fig 1.1

SOFTWARE ARCHITECTURE (1)

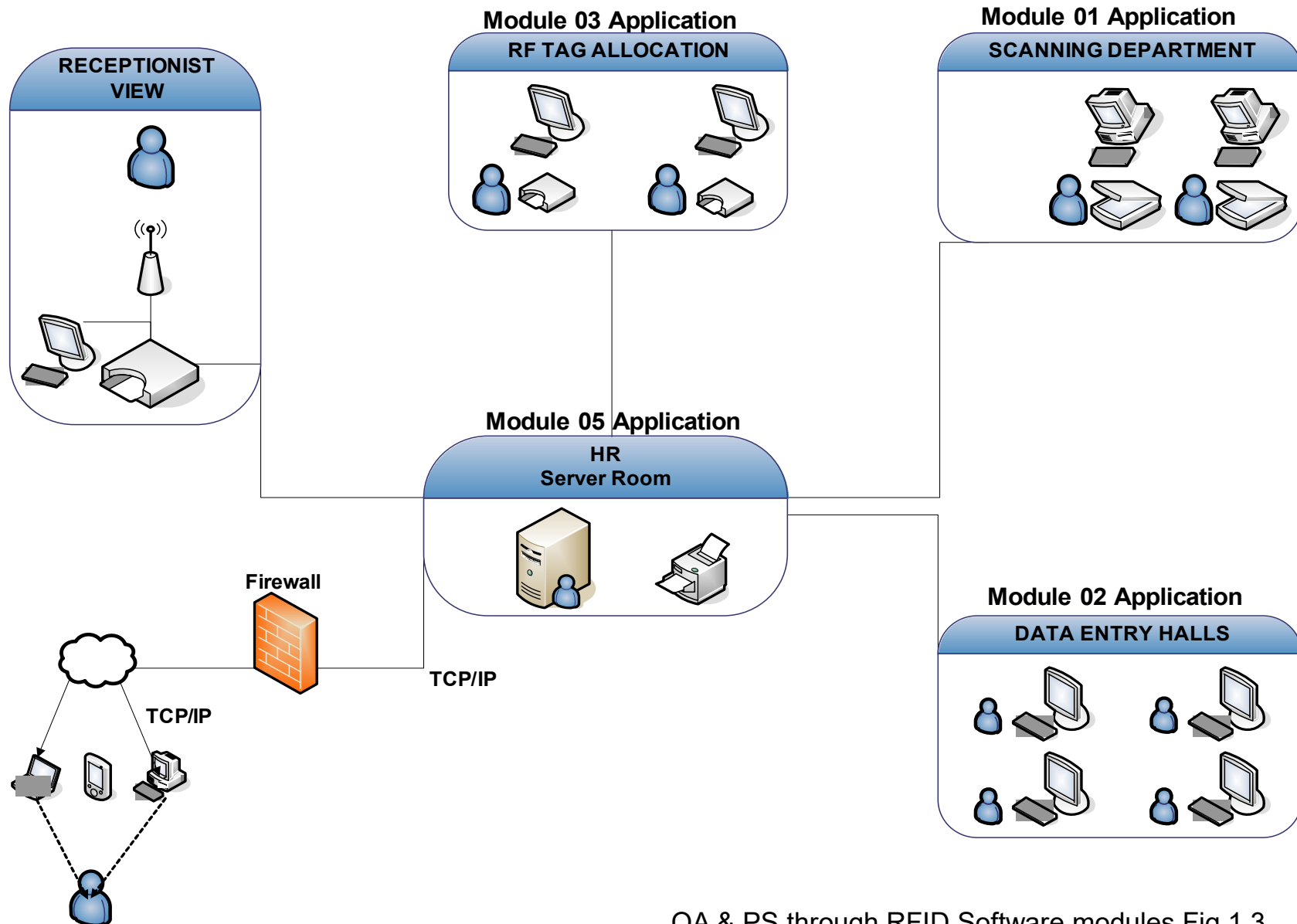


OA & PS through RFID Architecture Fig 1.2

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SOFTWARE MODULES



OA & PS through RFID Software modules Fig 1.3

SOFTWARE MODULES

- **Document scanning**
- **Data entry operation**
- **RF Tag allocation**
- **Reception**
- **Administration**
- **Web connective**
- **Fault tolerance**
- **Reports generating**

SOFTWARE MODULES

- **Document scanning**
 - Employee (having 'Scanner' Status) Connect through Application to central Database.
 - Scanner Scanned Hard copy of Employees forms.
 - These forms are available for DEO phase.
 - Rejected forms from DEO can be view/deleted/re-stored by the Scanner.

SOFTWARE MODULES

- **Data entry operation**
- Employee (having 'Data Entry Operator' Status) Connect through Application to central Database.
- DEO start Data Entry of Available forms.
- In-case of bad scanned Image, DEO can reject it.
- Finally employee info available in the Database for further process.

SOFTWARE MODULES

- **RF Tag Allocation**
- Employee (having 'HR' Status) Connect through Application to central Database.
- HR start Data Entry Verification of Available forms.
- In-case of wrong data entry, HR can reject it for again Data entry.
- In-case of successfully data entry, HR can Assign RF Tag Id to this Employee.
- HR can also Encrypt RF Tag Id for future Enhancement purpose.

SOFTWARE MODULES

- **Reception**
- Employee (having 'Receptionist' Status) Connect through Application to central Database.
- He/She Initialize Particular RF Reader.
- In-case of successfully RF Reader Initialization. RF Reader starts its work.
- This application also tells (by Speaking) that Employee is going to be IN or OUT.

SOFTWARE MODULES

- **Administration**
- Employee (having 'Administrator' Status) Connect through Application to central Database.

Admin account can perform following work.

- **Scanning**
- **DEO**
- **RF Tag allocation**
- **Reception**
- **Employee Account management**
- **Assigning/upgrading/deleting job status.**
- **Department management**
- **Pay Scale management**
- **And many more etc.**

SOFTWARE MODULES

- **Web connective**

At any time or at the end of a month an Employee

1. Check his/her attendance.
2. Check his/her Pay details.

- **Fault tolerance**

In special case when an employee forgets/loose his Employee Card at home. Generally he would not be able to mark his attendance.

1. But in this reader an Employee can mark his attendance manually.
2. In case of Power outage. Reader can contain UPTO 8000 RF Entries with date time. Those further can be downloaded from the Receptionist account.

- **Reports generating**

The proposed system provides the following types of reports. The user will be asked to select a choice among a number of choices. On the basis of the choice entered by the user, report will be generated.

Tools And Technologies

- **ORACLE 9i** as **RDBMS tool**
- **ASP. Net, C#** as **Deve tool**
- **RF Reader (13.56mghz) DLLS**
- **MS VISIO** as **Sys Analysis/Design tool**
- **Operating System** **Windows XP(Pro)**

Future Enhancements

- Upgrade RF hardware according to need.
- Tags can be upgraded to WORM/RW Tags.
- Encrypted data is available so Active tag can be work with this Software.
- Upgrading the application to its Accounts System
- The Website can contain more dynamic using the same database.
- Similarly desktop application can also be enhanced using the same database.
- New Version Web based can be implemented.

Thank You



Questions & Answers

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